Exam Date & Time: 27-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

DOWNSTREAM PROCESSES- II [BIO 3124]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.
Assume any missing data with proper reasoning.

1A) Two spent streams of Water-Acetic acid are mixed. First liquid stream (75 moles) with water (35% mole) and second stream (25 moles) with water (80% mole). The mixed solution was flashed at 26.66kPa such that final distillate mole fraction is 0.52. Compute the Nm^3 (at NTP) of vapor generated and composition of the residue liquid. Mole fraction data of water in VLE is as below. R=8.314 J/mol K (5)

x	0	0.3	0.454	0.633	0.769	0.857	0.927	0.967	1
y	0	0.4	0.573	0.733	0.839	0.899	0.949	0.978	1

1B) Describe the positive deviation from ideality in the case of VLE. Include azeotrope. (2)

1C) Define the following - Free Moisture, Equilibrium Moisture, Bound and Unbound Moisture. Use appropriate diagram. (3)

2A) At equilibrium, the concentration of water in vapour phase C^{eq} in kg/m^3 of air space and the amount of water (kg) adsorbed per kg of dry silica gel, m, are related by C^{eq}= 0.0667*m. It is required to maintain dry conditions in a room of air space 100 m^3 containing 2.2 kg of water vapour isothermally. **DO NOT USE GRAPH. DRAW THE FLOW DIAGRAM FOR EACH.**

i. If 10 kg of dry silica gel is kept in the room, compute the fraction of initial water remaining in the air space after a long time. (4)

ii. The same operation was done in two stages. Each time 5 kg of dry silica gel was used. What is the final absolute moisture content of the air in kg-moisture/kg dry-air? The density of air is $1.22kg/m^3$.

2B) Elucidate the plate theory of chromatography by taking up 3 equilibrium plates and a single component, K=3, feed concentration is 100 µmol and for 3 different time intervals. (4)

2C) What are the characteristics of an ideal adsorbent? (2)

3A) Define K_d in Size Exclusion Chromatography. Give reasons for three different values of K_d (K_d=0, K_d=1, 0< K_d < 1, K_d>1) in size Exclusion Chromatography (3)

3B) In a displacement chromatography for purification of β lactoglobulin in an ion exchange column using Tris Buffer, saccharin as displacer and NaCl as carrier buffer, trials were conducted under different conditions. The following problems different were noticed in different runs (i) Complete mixing of displacement zones (ii) Feed components are detached from the displacer List the reasons for the observed problems and the solutions for overcoming the problems (4)

3C) What are the reasons for the Tailing and Drift observed in a chromatograph? How can they be overcome? (3)

4A) Crystals of $Na_2CO_3 \cdot 10H_2O$ are dropped into a saturated solution of Na_2CO_3 in water at 100°C leading to dissolution of all the crystals and reprecipitation as $Na_2CO_3 \cdot H_2O$.What percent of the Na_2CO_3 in the $Na_2CO_3 \cdot H_2O$ is recovered in the precipitated solid? The precipitated solid is $Na_2CO_3 \cdot H_2O$. Data at 100°C: the saturated solution is 31.2% Na_2CO_3; molecular weight of Na_2CO_3 is 106. Assume 100 g $Na_2CO_3 \cdot 10H_2O$ is added to large pool of saturated solution. (4)

4B) Explain Langmuir isotherm and methodology to determine constants (3)

4C) Explain the Van Deemter relationship with example. (3)

5A) What are the four different stages in a crystallization process? Explain their significance. (2)

5B) How many grams of seed antibiotic with a size of 0.05 mm are required to produce 160 g/litre-solution, of a size of 1mm? Assume that the crystals are nearly uniform and have a density of 1.04 g/litre. Take one litre of solution as the basis. (3)

5C) In a drying experiment, a tray dryer consisting of single tray of 1 m^2 area is used to dry crystalline solids. The amount of dry solids is 4.12 kg. The following data has been collected. (5)

Time ,h	0	0.4	0.8	1	1.4	1.8	2.2	2.6	3	3.4	4.2	4.6	5	6	∞
Weight of wet substance kg	5.34	5.31	5.16	5.12	5.04	4.97	4.88	4.82	4.74	4.67	4.52	4.47	4.43	4.34	4.12

i. Compute rates of drying (kg/kg.m^2.h) at the individual times and plot the Rate (kg/kg.m^2.h) vs X (kg/kg); Use central difference method

ii. find the critical moisture content (use area under curve) and constant drying rate (average);

iii. If dry air is available at 40^0C with an absolute humidity of 0.01 kg water/kg dry air, and the dryer is maintained at 90^0C, calculate the amount of air required in the first two hours, assume that air is heated to 90^0C and the dry air leaves the dryer at 90^0C with 5% saturation. Vapor pressure of water at 90^0C is 525 mm Hg.

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